
 $df.shape = (150, 5)$
 Dimensionality reduction

① Feature selection:

SL SW PL PW flower

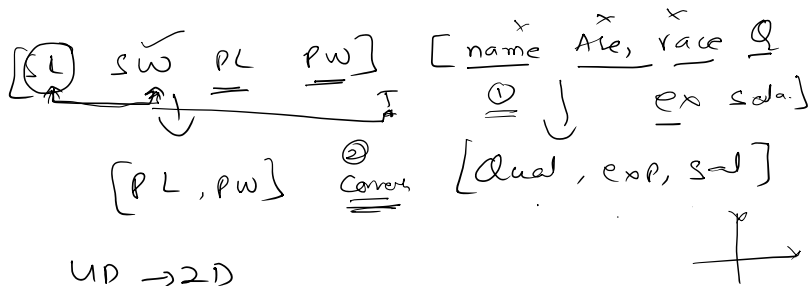
$x_1 = SL$

$x_2 = SW$

$x_3 = PL$

$x_4 = PW$

From existing columns choose only the most useful columns


 $UD \rightarrow 2D$

Selected only useful original features
Dimensionality reduction

① visualization →

② Compression →

③ Speed →

④ Noise removal →

⑤ Removing redundancy →

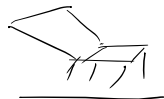
⑥ Finding hidden structure →

② Feature extraction:

create new features by combining the original features

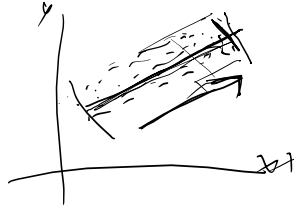
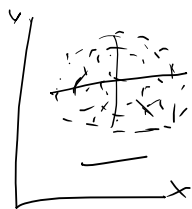
PC1, PC2

PCA (Principal Component Analysis)



② Eigen value

③ eigenvectors



- PCA →
- ① Center the data
 - ② Compute covariance matrix
 - ③ Compute eigen values and vectors
 - ④ project data onto eigenvectors

	H	W	BMI
0	126	78	49
1	128	180	48
2	128	82	50
3	130	82	48
4	130	84	49

$$\begin{aligned}
 H_c &= 126 - 128.4 = -2.4 \\
 W_c &= 78 - 128.4 = -50.4 \\
 W_c &= 180 - 128.4 = 51.6 \\
 W_c &= 82 - 128.4 = -46.4 \\
 W_c &= 82 - 128.4 = -46.4 \\
 W_c &= 84 - 128.4 = -44.4 \\
 BMI_c &= 49 - 48.8 = 0.2 \\
 BMI_c &= 48 - 48.8 = -0.8 \\
 BMI_c &= 50 - 48.8 = 1.2 \\
 BMI_c &= 48 - 48.8 = -0.8 \\
 BMI_c &= 49 - 48.8 = 0.2
 \end{aligned}$$

$$\bar{H} = \frac{126 + 128 + 128 + 130 + 130}{5} = 128.4$$

$$\bar{W} = 128.4$$

$$\bar{BMI} = 48.8$$

$$\text{mean value} = \begin{pmatrix} 128.4 \\ 128.4 \\ 48.8 \end{pmatrix}$$

$$48.8 \quad 48.8$$

$$\begin{matrix} H & W & BMI \\ X & \begin{pmatrix} H_c & W_c & BMI_c \\ -2.4 & -50.4 & 0.2 \\ -50.4 & 51.6 & -0.8 \\ -46.4 & -46.4 & 1.2 \\ -46.4 & -44.4 & -0.8 \\ 0.2 & -0.8 & 0.2 \end{pmatrix} \end{matrix}$$

$$\begin{aligned}
 78 - 81.2 \\
 80 - 81.2 \\
 81 - 81.2 \\
 82 - 81.2 \\
 84 - 81.2 \\
 \hline
 49 - 48.8 = 0.2 \\
 48 - 48.8 = -0.8 \\
 50 - 48.8 = 1.2 \\
 48 - 48.8 = -0.8 \\
 49 - 48.8 = 0.2
 \end{aligned}$$

$$(\text{cov}) \underline{S} = \frac{1}{n-1} X^T \cdot X \rightarrow$$

$$|\underline{S} - \lambda \underline{I}| = 0 \Rightarrow \text{eigen values}$$

$$\underline{S} = \begin{bmatrix} 2.8 & 3.4 & -0.4 \\ \dots & \dots & \dots \end{bmatrix}$$

$$\underline{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\underline{I}_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$I = 0.1$

$$T_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|S - IT| = 0$$

$$\left| \begin{pmatrix} 2.8 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.3 \\ -0.4 & 0.3 & 0.7 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right| = 0$$

$$\begin{pmatrix} 2.4 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.2 \\ -0.4 & 0.2 & 0.8 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} = 0 \quad \left| \begin{smallmatrix} a & b \\ c & d \end{smallmatrix} \right| = \underline{\underline{ad-bc}}$$

$$\begin{pmatrix} 2.8 - x & 1.4 & -0.4 \\ 1.4 & 5.2 - x & 0.2 \\ 0.4 & 0.2 & 0.7 - x \end{pmatrix} = 0$$

$$\lambda_1 \quad \lambda_2 \quad \lambda_3$$

$$\underline{a(ei - fh)} - \underline{b(di - eh)} + c(dh - eg)$$

$$\begin{aligned} & \underline{(2.8 - \lambda) \left((5.2 - \lambda) \times (0.3 - \lambda) \right) - (0.3) \times 0.3} \\ & \quad - 0.4 \left((3.4) \times 0.2 + 0.3 \times (6.4) \right) \\ & \quad - 0.4 \left(3.4 \times 0.2 + (5.2 - \lambda) \times (-0.4) \right) = 0 \\ & \quad - \lambda^3 + 8.7\lambda^2 - 8.5\lambda + 0.199 = 0 \quad \quad \quad \lambda \neq \sqrt{5.2 \text{ var}} \\ & \quad \text{eigenvalues } \lambda_1, \lambda_2, \lambda_3 \end{aligned}$$

Eigen vectors:

$$\begin{aligned}\lambda_1 &= 7.60 \\ \lambda_2 &= 1.06 \\ \lambda_3 &= 0.02\end{aligned}$$

$$(S - \lambda I) \underline{V} = 0$$

$$v_1 = \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$$

$$V_L = \textcircled{V_3}$$

$$\begin{pmatrix} 2.8 & 2.4 & -0.4 \\ 2.4 & 5.2 & 0.2 \\ -0.4 & 0.2 & 0.4 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 0$$

$$\begin{pmatrix} 2.8 & 3.4 & -0.4 \\ 3.4 & 5.2 & 0.3 \\ -0.4 & 0.3 & 6.9 \end{pmatrix} - \begin{pmatrix} 7.6 & 0 & 0 \\ 0 & 7.6 & 0 \\ 0 & 0 & 7.6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{pmatrix} 2.8 - 7.6 & 3.4 & -0.4 \\ 3.4 & 5.2 - 7.6 & 0.3 \\ -0.4 & 0.3 & 0.3 - 7.6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\begin{pmatrix} -4.8 & 3.4 & -0.4 \\ 3.4 & -2.4 & 0.3 \\ -0.4 & 0.3 & -6.9 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$-4.8x + 3.4y - 0.4z = 0 \quad \text{--- (1)}$$

$$3.4x - 2.4y + 0.3z = 0 \quad \text{--- (2)}$$

$$-0.4x + 0.3y - 6.9z = 0 \quad \text{--- (3)}$$

$$z = 1$$

$$-4.8x + 3.4y - 0.4 = 0 \Rightarrow 0.4$$

$$3.4x - 2.4y + 0.3 = 0 \Rightarrow -0.3 \quad \lambda_1 = 7.6$$

$$-0.4x + 0.3y - 6.9 = 0$$

$$x = y =$$

centered the data

cov matrix

eigen values, eigenvectors

$$\begin{pmatrix} PC1 \\ PC2 \\ PC3 \end{pmatrix}$$

$$v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1.2 \\ 0.2 \\ 1 \end{pmatrix}$$

$$|v| = \sqrt{(-1.2)^2 + (0.2)^2 + 1^2} = 0.8$$

$$v = \frac{1}{0.8} \begin{pmatrix} -1.2 \\ 0.2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1.2/0.8 \\ 0.2/0.8 \\ 1/0.8 \end{pmatrix} = \begin{pmatrix} -1.5 \\ 0.25 \\ 1.25 \end{pmatrix} = v_1$$

$$v = \frac{1}{0.4} \begin{pmatrix} \frac{-1.2}{0.7} \\ 1 \end{pmatrix} = \begin{pmatrix} -1.2/0.4 \\ 0.7/0.4 \\ 1/0.4 \end{pmatrix} = \begin{pmatrix} -3 \\ 1.75 \\ 2.5 \end{pmatrix} \quad \text{--- } v_1$$

$$\lambda_1 = -2.6 \quad \lambda_2 = 1.2 \quad \lambda_3 = 0.007$$

$$(A - \lambda I)(v) = 0$$

$$(A - -2.6 I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad \text{--- } v_1$$

$$(A - 1.2 I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad \text{--- } v_2$$

$$(A - 0.007 I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad \text{--- } v_3$$

$$\lambda_1 = -2.6 \quad \lambda_2 = 1.2 \quad \lambda_3 = 0.007$$

$$P_{total} = \lambda_1 + \lambda_2 + \lambda_3 = 8.8007$$

$$PC_1 = \frac{(8.8007)}{-2.6} \times 100 = 86.6\%$$

$$PC_2 = \frac{1.2}{8.8007} \times 100 = 13.7\%$$

$$PC_3 = 0.007 / 8.8007 \times 100 = 0.4\%$$

$$PC_1, PC_2 \quad \text{--- } PC_1 \quad \text{--- } \begin{pmatrix} A & W & Bmg. \\ 1 & 1 & 1 \end{pmatrix}$$

$$\begin{array}{c|c} PC_1 & PC_2 \\ \hline & \end{array}$$

$$v_1 \begin{pmatrix} 0.5 & 0.2 & 0.3 \\ 0.5 & 0.2 & 0.1 \\ 0.2 & 0.3 & 0.4 \end{pmatrix}$$

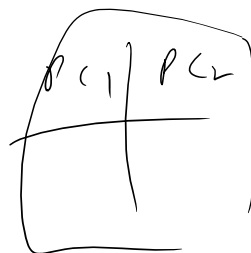
H_c	W_c	$bias_c$	PC_1	PC_2	PC_3
0.2	-	=	-0.1	-0.1	0.2

① Center the H W $bias$

② Covariance

③ eigenvalues & eigenvectors

④ PC_1 PC_2 PC_3



PCA is unsupervised algorithm ~~XXXXX~~
 that means PCA does not ask
 which feature help predict the target

in the features which direction has maximum spread / variance

$$X_c = X - \bar{X}$$

$$S = \frac{1}{n-1} X_c^T \cdot X_c$$

Now we have target

① PCA

① find new axes where the data has maximum variance ✓

H W $bias$
 $\downarrow$
 PC_1 PC_2

tq, CNN, SVM, Clus, U.S. below

② tSNE

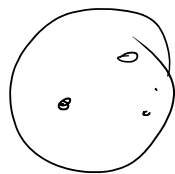
Isometric mapping

non-linear dimensional method

Swiss roll data

PCA ✗

③ LLE → locally linear Embedding
non-linear



local structure

④ t-SNE

t-distributed stochastic Neighbor Embedding

non-linear visualization

2D or 3D



from ✗ loc